

Abstract

- Our project enables secure online medicine ordering with prescription verification.
- The system provides secure prescription validation and role-based access making it more organized and secure.

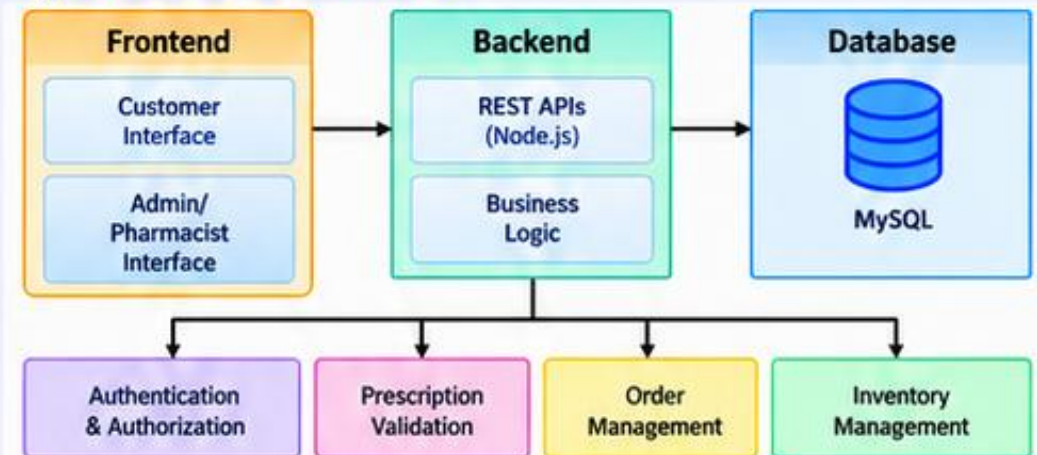
Problem Statement & Objectives

- Problem: Existing online pharmacies lack secure prescription verification and centralized management of pharmacy records.
- Objective 1: Enable secure online medicine orders
- Objective 2: Verify Prescriptions before purchase
- Objective 3: Centralize medicines, orders and records
- Scope: Customers, Pharmacists, and Admins

Tools & Technologies

- Front-End: [React / HTML-CSS-JS]
- Back-End: [Node.js / Spring Boot / Django] + REST APIs
- Database: [MySQL]
- Tools: [Git & GitHub, Postman, VS Code, Draw.io]

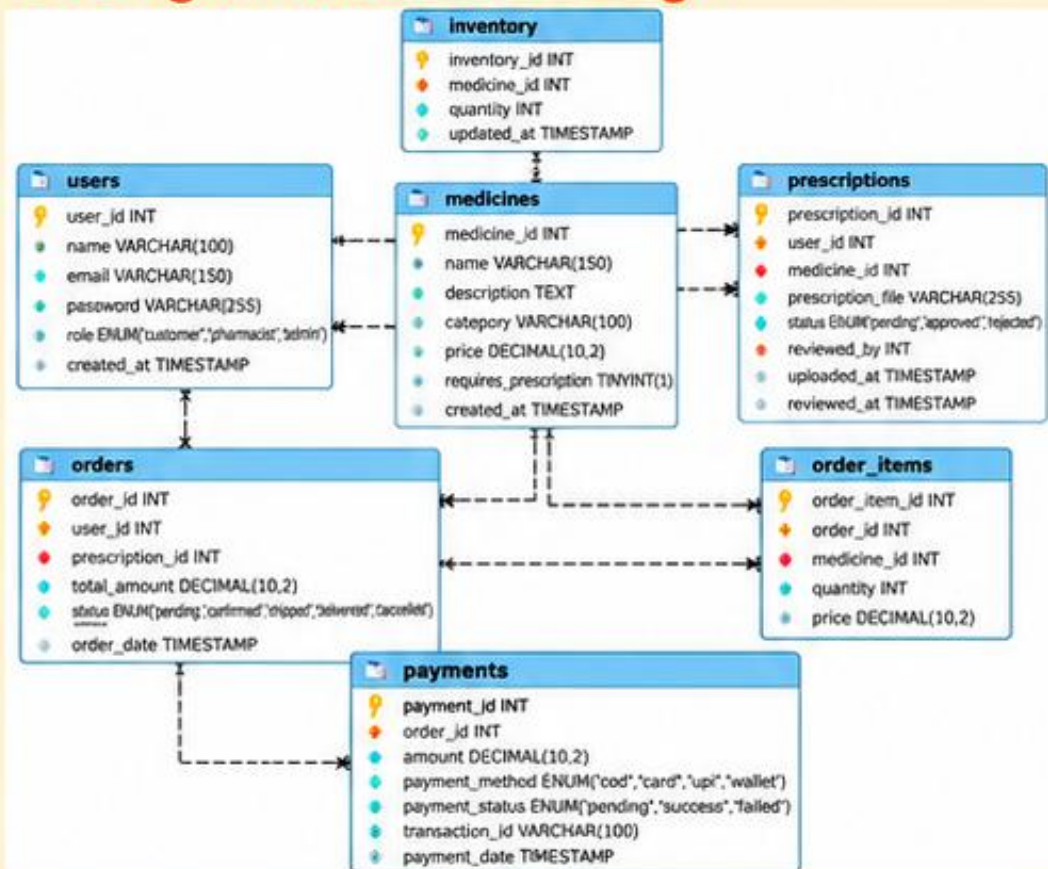
System Architecture



Existing vs Proposed System

Feature	Existing System	Proposed System
Prescription	Manual verification	Online upload & validation
Records	Disconnected records	Centralized MySQL database
Access	Limited role separation	Role-based access
Ordering	Basic medicine ordering	Secure medicine ordering
Limitation / Gap	Slow and less integrated	Secure and integrated system

ER Diagram & Database Design



Feasibility Study

- Technical:** Uses React.js, Node.js, Express.js, and MySQL.
- Economic:** Uses open-source and freely available tools.
- Operational:** Easy-to-use system for customers, pharmacists, and administrators.
- Schedule:** Can be completed within the semester timeline.

Implementation Screenshots



Results & Testing

- CRUD operations tested successfully for core database records.
- Prescription validation and order processing were tested for correctness.
- Database relationships and input validation were verified.

Conclusion

- Provides secure and efficient online medicine ordering with prescription validation.
- Meets the objectives through centralized database management and role-based access.

Future Scope

- Add online payment and delivery tracking.
- Improve prescription verification and expand pharmacy services.